DAVID ABELLA BUJALANCE

Researcher in Computational Social Science

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SUMMARY

PhD in Physics of Complex Systems (IFISC, 2024) and researcher in Computational Social Science at Eurecat. I apply complex systems physics to social and urban challenges, including housing market dynamics, urban segregation, collective behavior and online platforms such as Wikipedia, combining data-driven tools, network science and statistical physics.

SKILLS

Languages: Català, Castellano and English

Coding Languages: Python, Julia, Fortran, SQL and HTML.

Tools: Linux, Git/GitHub, Pandas, GeoPandas, PyTorch, TensorFlow, Parallel Computing and Inkscape.

EXPERIENCE

01/2025 – Present **Researcher in Computational Social Science** — **Eurecat, Centre Tecnològic de Catalunya**
- DEM-Debate: Wikipedia editing and content moderation during the 2024 European elections.
- AI4Bus: AI forecasting of passenger flows for Barcelona's metropolitan bus network (with AMB).
- deployEMDS: deployment of the common European Mobility Data Space (Digital Europe Programme).

2023 – 2024 **Predoctoral Researcher** — **Instituto de Fisica Interdisciplinar y Sistemas Complejos (IFISC - UIB/CSIC)**
- Real estate data analysis and modelling via geolocalization techniques and clustering methods.

09 – 12/2023 **iMOVE Grant – Research stay** — **School of Mathematical Sciences, QMUL, London**
- New tools to assess social segregation from first principles, applied to many segregation datasets.

2022 – 2023 **SOIB Grant - Young Research Talent** — **Universitat de les Illes Balears (UIB)**
- Agent based modelling of socio-technical systems, focused on complex contagion.

2021 – 2022 **Predoctoral Researcher** — **Instituto de Fisica Interdisciplinar y Sistemas Complejos (IFISC - UIB/CSIC)**
- Study of segregation with aging dynamics. Phase diagram analysis and modelling.

EDUCATION

10/2026 **Agile training** — **Kairós (at Eurecat)**
Agile methodologies for project management and team collaboration.

10/2021 - 10/2024 **Ph.D in Physics of Complex Systems** — **Instituto de Fisica Interdisciplinar y Sistemas Complejos (IFISC - UIB/CSIC)**
Thesis: *Aging and memory effects in social and economic dynamics*. Awarded the doctoral thesis prize of the UIB Chair of Urban Studies (2024).

9/2019 - 12/2020 **Master in Physics of Complex Systems** — **Instituto de Fisica Interdisciplinar y Sistemas Complejos (IFISC - UIB/CSIC)**
Emergent properties of complex systems, stochastic processes and data-driven modeling.

9/2015 - 6/2019 **Bachelor degree in Physics** — **Universitat de Barcelona**
With Minor in Fundamental Physics.

PUBLICATIONS

2026 **Computational analysis of Wikipedia editorial activity during the 2024 European Parliament election** — **Software & data**
Reproducible pipeline and data on editing, reverts and moderation in 31 Wikipedia language editions.

2025 **Exploring the spatial segmentation of housing markets from online listings** — **EPJ Data Science**
Multipartite networks reveal the spatial segmentation of real estate markets across countries.

2024 **Ordering dynamics and aging in the symmetrical threshold model** — **New J. Phys.**
Phase diagram of a symmetric Granovetter–Watts model and how aging alters its phases.

2023 **Aging in binary-state models: The Threshold model for complex contagion** — **Phys. Rev. E**
Aging slows down complex contagion, yielding stretched exponential or power-law dynamics.

2023 **Many-body contributions in water nanoclusters** — **ACS Nano**
The first coordination shell captures the energy minimum of water nanoclusters within 5%.

2022 **Aging effects in Schelling segregation model** — **Scientific Reports**
Attachment to the current location changes both the statics and dynamics of segregation.